

# Forest Fire and Smoke Detection Using Yolov8n

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**Abstract**—Early detection of fire and smoke is critical for mitigating the catastrophic impact of fire-related incidents on life, property, and the environment. In this paper, we present a fire and smoke detection framework. We conduct a comprehensive analysis of a specialized dataset, performing rigorous data integrity checks and class distribution assessments on over 21,000 annotated instances to ensure yolov8n model reliability. Our methodology involves the extraction of high-dimensional visual features to distinguish between volatile smoke patterns and active fire zones. Experimental results demonstrate that the proposed system achieves high precision and recall, effectively identifying hazards within milliseconds. However, the study also highlights the challenges of environmental noise. Our findings suggest that the integration of optimized YOLO models into existing surveillance infrastructures provides a scalable and efficient solution for hazard monitoring.

**Index Terms**—Computer Vision, Fire Detection, YOLO Architecture, Deep Learning, Disaster Mitigation.

## I. INTRODUCTION

Forest fires are one of the most critical environmental challenges, causing extensive damage to ecosystems, wildlife, and human infrastructure. The increasing frequency of wildfires due to climate change, rising temperatures, and prolonged dry conditions has made early detection an essential requirement for effective disaster management. Rapid identification of fire and smoke at an early stage can significantly reduce the spread of fire and minimize ecological and economic losses.

Some fire detection techniques such as satellite monitoring, watchtower surveillance, and sensor-based systems, suffer from several limitations. Satellite-based systems often experience delayed detection due to low temporal resolution and obstruction caused by clouds or atmospheric conditions. Manual monitoring through watchtowers is labor-intensive and prone to human error, while sensor-based approaches require large-scale deployment and high maintenance costs. These limitations highlight the need for an automated, scalable, and real-time detection system.

With the advancement of deep learning and computer vision, image-based fire detection has emerged as a promising alternative. Convolutional Neural Networks (CNNs) have shown remarkable capabilities in extracting meaningful features from visual data, enabling accurate detection and classification of objects. Among various object detection models, the YOLO (You Only Look Once) family is widely recognized for its detection capability and high efficiency. In this work, we propose a forest fire and smoke detection system using the YOLOv8n model, which is a lightweight and computationally

efficient variant of the YOLOv8 architecture. The choice of YOLOv8n is motivated by its ability to provide a balance between detection accuracy and inference speed.

The implementation of the proposed system involves multiple stages, including dataset selection and processing, model training, validation, and testing. A wildfire dataset consisting of fire and smoke images is utilized, and annotations are created using bounding boxes in YOLO format. The dataset is divided into training, validation, and testing sets to ensure proper evaluation of model performance. Transfer learning is employed using pre-trained YOLOv8n weights, allowing the model to learn domain-specific features efficiently while reducing training time.

During training, the model learns to detect two primary classes: fire and smoke. Special attention is given to ensuring that bounding boxes accurately capture only the relevant regions of fire and smoke, avoiding unnecessary background noise. The trained model is then evaluated using standard performance metrics such as precision, recall, and F1-score, which provide insights into detection accuracy and reliability. The trained YOLOv8n model is integrated with an image processing pipeline using computer vision libraries, enabling it to process frames continuously and generate bounding box predictions for fire and smoke.

Despite the effectiveness of the proposed approach, certain challenges are observed during implementation. Variations in environmental conditions, such as lighting changes, fog, and visually similar patterns like clouds or dust, can affect detection accuracy. The main contributions of this work include the development of a lightweight and efficient fire and smoke detection system using YOLOv8n, implementation of a complete training and testing pipeline with annotated datasets. The study also provides an analysis of performance detection accuracy.

## II. PROPOSED METHODOLOGY

### A. Overview of the Proposed System

The proposed system is designed for detection of forest fire and smoke using a deep learning-based object detection framework. The system utilizes the YOLOv8 model to analyze visual data obtained from images. The primary objective of the system is to enable early detection of fire hazards by identifying both fire and smoke regions with high accuracy and minimal delay.

The overall workflow of the system consists of a sequence of interconnected stages, including input preprocessing, hi-

erarchical feature extraction, feature enhancement, multi-scale feature aggregation, object detection. Initially, the input image is standardized to a fixed resolution of  $640 \times 640$  pixels to ensure uniformity during model inference. This preprocessing step reduces computational complexity while preserving important spatial information required for detection. The processed input is then passed through a deep neural network that progressively transforms raw pixel data into meaningful feature representations.

The system is designed to operate efficiently in environments, where continuous frame processing is required. By leveraging a single-stage detection mechanism, the model simultaneously performs localization and classification, eliminating the need for separate region proposal steps. This significantly reduces inference time and makes the system suitable for deployment in time-critical applications such as forest monitoring and disaster management.

### B. Detailed Methodology

The proposed methodology is based on a structured deep learning pipeline that transforms input visual data into detected fire and smoke regions through a series of computational stages. Each stage is designed to enhance the representation of relevant features while suppressing noise and irrelevant information, thereby improving detection accuracy.

The process begins with input acquisition, where an image is captured and represented as :

$$I \in R^{H \times W \times 3} \quad (1)$$

where H and W denote the height and width of the image, and the third dimension corresponds to the RGB color channels. The image is resized to a fixed dimension of  $640 \times 640$  pixels and normalized to ensure numerical stability during processing. This step ensures that the model receives consistent input regardless of the original resolution of the data.

Following preprocessing, the input is subjected to feature extraction, where convolutional operations are applied to identify patterns within the image. The fundamental operation of convolution can be expressed as:

$$F(x, y) = \sum_{i=0}^k \sum_{j=0}^k I(x+i, y+j) \cdot K(i, j) \quad (2)$$

where I represents the input image, K is the convolution kernel, and F(x,y) is the resulting feature map. This operation allows the model to detect local patterns such as edges, textures, which are crucial for identifying fire and smoke.

As the data propagates through deeper layers, the receptive field of the network increases, enabling it to capture more complex and abstract features. Low-level layers focus on basic visual attributes such as color gradients and edges, which are particularly useful for detecting flame boundaries. Intermediate layers capture textures and irregular patterns associated with smoke dispersion. Higher-level layers extract semantic

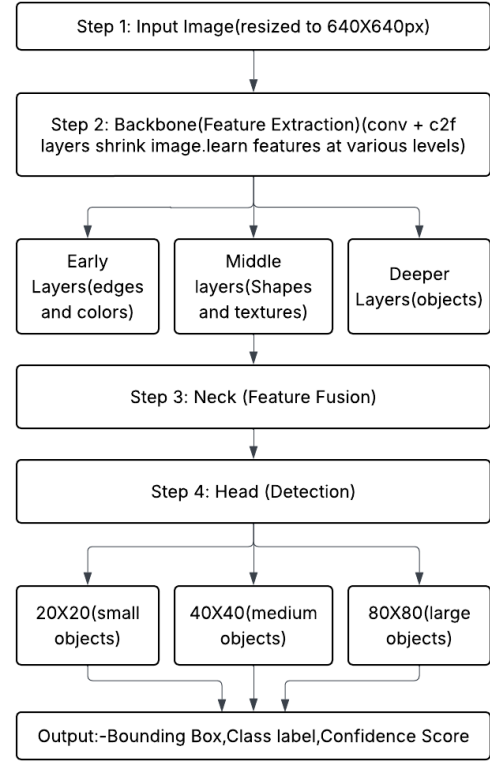


Fig. 1. WorkFlow of the Model

information, allowing the model to distinguish between fire, smoke, and background elements.

To further enhance feature representation, multi-scale contextual processing is applied. This enables the model to analyze the image at different resolutions simultaneously, ensuring that both small and large objects can be detected effectively. In forest environments, this capability is essential because smoke often appears as a small, diffused region, whereas fire may occupy larger and more prominent areas.

The feature maps generated at different levels are then aggregated using a multi-scale feature fusion strategy. This process combines high-resolution spatial information with low-resolution semantic information, improving both localization and classification performance. The fusion mechanism ensures that fine-grained details are preserved while maintaining a strong understanding of the overall scene context.

The fused feature representations are then forwarded to the detection stage, where the model performs simultaneous object localization and classification. In this stage, the processed feature maps are interpreted in a grid-based manner, where the image is conceptually divided into an  $S \times S$  grid. Each grid cell is responsible for predicting objects whose centers fall within its spatial region. For each grid cell, the model predicts multiple bounding boxes along with associated confidence scores and class probabilities.

Each detected object is represented by a bounding box, a

corresponding class label, and an associated confidence score. These outputs collectively describe the spatial location and category of the detected regions within the input image. Each bounding box is defined by four parameters, namely the center coordinates (x,y), width w, and height h, which are normalized with respect to the image dimensions. These normalized values are then scaled back to the original image resolution to accurately map the detected regions onto the input frame. The class label assigned to each bounding box corresponds to one of the predefined categories, which in this case include fire and smoke. The confidence score indicates the reliability of the prediction and is used to filter out low-confidence detections.

Once the detections are obtained, they are visualized by overlaying bounding boxes on the original image. Each bounding box is accompanied by a label indicating the detected class and its corresponding confidence value. This visualization step provides an intuitive understanding of the detection results and allows for real-time monitoring of forest fire conditions.

Additionally, a thresholding mechanism is applied to confidence scores to ensure that only reliable detections are considered. Predictions with confidence values below a predefined threshold are discarded to reduce false positives. This step improves the robustness of the system, particularly in challenging environmental conditions such as fog, clouds, or varying lighting.

### C. Model Architecture

The proposed forest fire detection system is built upon a deep learning-based object detection architecture that follows a hierarchical design for efficient feature extraction, multi-scale representation, and accurate prediction. The model processes the input image through a sequence of transformations that progressively convert raw pixel information into meaningful semantic representations. The overall architecture consists of three functional stages: feature extraction, feature aggregation, and prediction generation, which collectively enable robust detection of fire and smoke in complex environments. The model follows a hierarchical processing pipeline in which the input image is progressively transformed into high-level semantic representations through a sequence of feature extraction, contextual enhancement, multi-scale fusion, and prediction stages. This structured design enables the model to detect both fire and smoke effectively, even under complex environmental conditions.

The input to the model is a three-channel RGB image represented as in eqn. (1). The initial stage of processing involves convolutional transformations, where local spatial features are extracted using learnable filters. A convolution operation computes a weighted sum of pixel intensities within a local neighborhood and is mathematically defined in eqn. (2). This operation is fundamental to the network as it enables the detection of primitive visual patterns such as edges, gradients, and color transitions, which are essential for identifying fire boundaries and smoke diffusion.

As the input propagates through successive convolutional layers, the spatial dimensions of the feature maps are reduced

while the number of channels increases. This transformation is governed by the relation:

$$H' = \frac{H - K + 2P}{S} + 1, \quad W' = \frac{W - K + 2P}{S} + 1 \quad (3)$$

where S is the stride and P is the padding applied during convolution. This downsampling process allows the model to capture increasingly abstract features while maintaining computational efficiency.

**1) Backbone: Hierarchical Feature Learning:** The backbone serves as the primary feature extractor and is composed of multiple stacked convolutional and feature refinement blocks. In the initial layers, the network focuses on capturing low-level features such as edges and color gradients, which are particularly useful for identifying flame contours. As the depth increases, intermediate layers extract texture-based patterns that correspond to smoke dispersion and irregular intensity variations. In deeper layers, the network captures high-level semantic features, enabling it to distinguish between fire, smoke, and background elements.

A key characteristic of the backbone is its ability to generate feature maps at multiple resolutions. Specifically, feature maps of sizes 80×80, 40×40, and 20×20 are produced, each encoding information at different levels of abstraction. These multi-scale representations form the foundation for robust object detection, as they allow the model to detect both small and large objects effectively.

To further enhance feature representation, the backbone incorporates shortcut connections that facilitate efficient gradient flow during training. These connections allow information to bypass certain layers, reducing the risk of vanishing gradients and improving convergence stability. As a result, the model is able to learn deeper representations without degradation in performance. These convolutional layers are organized into advanced feature fusion modules known as C2f blocks, which improve gradient propagation and enhance feature reuse while maintaining computational efficiency.

#### SPPF: Multi-Scale Contextual Aggregation

At the terminal stage of the backbone, a Spatial Pyramid Pooling Fast (SPPF) module is employed to expand the receptive field of the network. This module performs repeated pooling operations with different receptive field sizes, enabling the model to capture both local and global contextual information.

The pooling operation can be expressed as:

$$P(x, y) = \max_{(i,j) \in R} F(x + i, y + j) \quad (4)$$

where R represents the pooling region. By applying pooling at multiple scales, the SPPF module enables the model to aggregate information from different spatial extents, which is crucial for detecting objects with varying sizes and shapes. In the context of forest fire detection, this allows the model to identify both compact fire regions and widely dispersed smoke patterns.

2) **Neck: Bidirectional Feature Fusion:** The neck component is responsible for integrating feature maps from different stages of the backbone to create a unified multi-scale representation. This is achieved through a combination of upsampling, downsampling, and concatenation operations, forming a bidirectional information flow.

Upsampling increases the spatial resolution of feature maps, allowing high-level semantic information to be projected onto finer spatial grids. This operation can be expressed as:

$$F_{up}(x, y) = F(sx, sy) \quad (5)$$

where  $s$  is the scaling factor. Conversely, downsampling reduces spatial resolution while preserving semantic richness.

Feature concatenation combines multiple feature maps along the channel dimension:

$$F_{concat} = [F_1 \oplus F_2] \quad (6)$$

This fusion mechanism ensures that detailed spatial information from shallow layers is combined with abstract semantic information from deeper layers. As a result, the model achieves improved localization accuracy while maintaining strong classification performance.

The neck ultimately produces refined feature maps at multiple scales, each optimized for detecting objects of specific sizes.

3) **Head: Multi-Scale Prediction Mechanism:** The detection head operates on the fused feature maps and generates predictions at multiple resolutions. Each resolution corresponds to a different detection scale, enabling the model to handle objects of varying sizes within the same image.

The detection process is formulated by dividing the image into a grid of size  $S \times S$ , where each grid cell is responsible for predicting multiple bounding boxes. Each bounding box prediction consists of spatial coordinates  $(x, y, w, h)$  and a confidence score. The confidence score is defined as:

$$\text{Confidence} = P(\text{Object}) \times \text{IoU} \quad (7)$$

where IoU measures the overlap between predicted and ground truth bounding boxes:

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (8)$$

This formulation ensures that both localization accuracy and object presence are considered during prediction.

The localization loss ensures accurate bounding box regression, the classification loss ensures correct labeling of fire and smoke, and the confidence loss penalizes incorrect predictions. This multi-objective optimization enables the model to achieve a balance between detection accuracy and robustness.

**Post-Processing and Final Output**

After generating predictions, the model applies Non-Maximum Suppression to remove redundant bounding boxes. This process selects the bounding box with the highest confidence score and suppresses overlapping boxes that exceed a predefined IoU threshold:

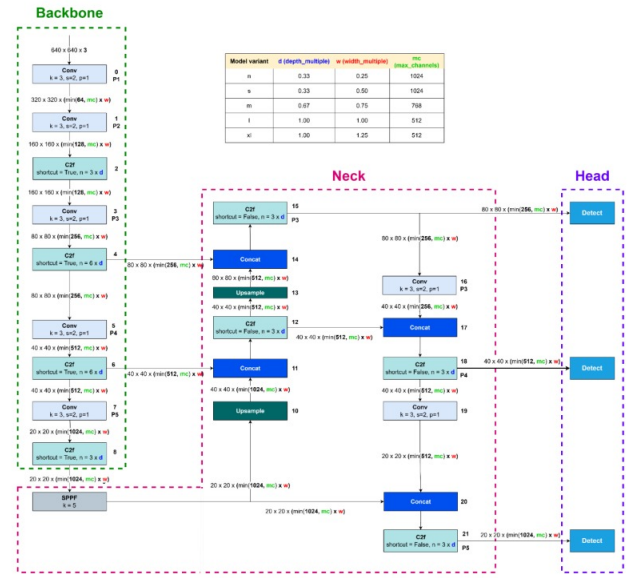


Figure 1: YOLOv8 Architecture (Adapted from [11], [12])

Fig. 2. Architecture of the YOLOv8n Model

The final output consists of a set of bounding boxes, each associated with a class label and confidence score. These detections are mapped onto the original image, enabling real-time visualization of fire and smoke regions.

#### D. Dataset Used

The Smoke-Fire-Detection-YOLO dataset, a publicly available dataset specifically designed for object detection tasks involving fire and smoke. The dataset is organized in the YOLO annotation format, allowing seamless integration with modern object detection frameworks such as Ultralytics YOLOv8. It is derived from the D-Fire dataset and has been refined to improve annotation quality, structural consistency, and usability for deep learning applications. The dataset contains a diverse collection of real-world images, making it highly suitable for developing robust fire detection systems applicable to both forest and non-forest environments.

The dataset comprises approximately 21,000 images annotated across two primary object classes, namely smoke and fire. These classes are encoded numerically as class 0 for smoke and class 1 for fire. The dataset includes a balanced combination of positive and negative samples, which is essential for training an effective detection model. Specifically, the dataset contains 1,164 images depicting only fire, 5,867 images containing only smoke, and 4,658 images that include both fire and smoke simultaneously. In addition, there are 9,838 background images that do not contain any instances of fire or smoke. The inclusion of these negative samples plays a crucial role in improving the model's ability to differentiate between relevant and irrelevant visual features.

In terms of object annotations, the dataset includes 14,692 labeled instances of fire and 11,865 labeled instances of smoke. Each object within an image is annotated using the

YOLO format, where bounding boxes are defined by normalized coordinates representing the center position, width, and height of the object relative to the image dimensions. This normalized representation ensures compatibility across different image resolutions and simplifies preprocessing requirements. Multiple objects can be annotated within a single image, enabling the model to learn complex scenarios where fire and smoke coexist.

The dataset is organized into separate subsets for training, validation, and testing, ensuring a systematic approach to model development and evaluation. The training set is used to learn model parameters, the validation set is used to fine-tune hyperparameters and monitor performance during training, and the test set is reserved for final evaluation. This structured partitioning ensures that the model's performance is assessed in an unbiased manner and helps prevent overfitting.

A significant strength of the dataset lies in its diversity and realism. The images are captured from a wide range of environments, including forest landscapes, urban settings, industrial areas, and indoor scenes. This diversity enhances the generalization capability of the trained model, enabling it to perform effectively across different real-world scenarios. Furthermore, the dataset incorporates various environmental challenges such as low-light conditions, haze, fog, varying smoke densities, and different fire intensities. These variations introduce complexity into the dataset, making it highly suitable for training robust detection systems capable of handling real-world uncertainties.

Prior to training, several preprocessing steps were applied to ensure data consistency and compatibility with the model. All images were resized to a fixed resolution, typically  $640 \times 640$  pixels, in accordance with the input requirements of YOLOv8. Pixel values were normalized, and annotation files were verified to ensure correctness and alignment with corresponding images. To further enhance model performance and reduce overfitting.

The dataset is particularly relevant for forest fire detection applications due to the presence of early-stage smoke patterns, which are critical indicators of potential fire hazards. The inclusion of images with varying fire intensities further enhances the model's ability to detect fires at different stages of development. Moreover, the availability of background images without fire or smoke helps the model distinguish between actual fire events and visually similar patterns, thereby reducing false alarms in practical deployments.

Despite its advantages, the dataset has certain limitations that must be considered. The presence of a large number of background images introduces class imbalance, which may affect model training if not properly addressed. Additionally, the dataset is not exclusively focused on forest environments, which may limit its specificity for certain applications. Visual similarities between smoke and environmental elements such as fog or clouds may also introduce ambiguity, potentially affecting detection accuracy under certain conditions.

### *E. Model Training*

The training process for the proposed fire and smoke detection system was carried out using the Ultralytics YOLOv8 nano model (YOLOv8n), selected for its lightweight architecture and suitability for real-time deployment in forest monitoring applications. The model was trained on the Smoke-Fire-Detection-YOLO dataset, which provides annotated images of fire and smoke across diverse environmental conditions, including forest regions, urban areas, and complex backgrounds.

The training strategy involved hyperparameter tuning across two configurations, specifically training the model for 50 epochs and 150 epochs, to evaluate the impact of training duration on detection performance. In both configurations, input images were resized to a resolution of  $640 \times 640$  pixels to standardize the input dimensions and optimize computational efficiency. A batch size of 16 was used to ensure stable gradient. The initial learning rate was set to 0.001, and optimization was performed using stochastic gradient descent with a momentum value of 0.937 and a weight decay of 0.0005, which helped in accelerating convergence while reducing the risk of overfitting.

During training, the model processed input images through successive convolutional and feature fusion layers, generating multi-scale feature representations. Predictions were evaluated using a composite loss function consisting of bounding box regression loss, classification loss, and Distribution Focal Loss. The bounding box loss ensured accurate localization of detected objects, while the classification loss improved the model's ability to correctly identify fire and smoke classes. The Distribution Focal Loss further refined the precision of bounding box predictions by modeling the probability distribution of object boundaries. Backpropagation was then applied to iteratively update the model weights, minimizing the overall loss and improving detection accuracy over time.

The initial training conducted for 50 epochs served as a baseline experiment, providing insight into the model's convergence behavior and initial learning capability. This configuration resulted in relatively fast training and moderate detection performance, suitable for preliminary evaluation. However, the extended training performed over 150 epochs allowed the model to learn more complex feature representations and significantly improved its ability to detect subtle smoke patterns and small fire regions. This is particularly critical in forest fire detection, where early-stage smoke identification can enable timely intervention and prevent large-scale damage. The longer training duration also contributed to better generalization and reduced false positives, especially in challenging scenarios involving visually similar elements such as clouds, fog, or dust.

The overall training process was conducted in a GPU-enabled computational environment using Python, ensuring efficient handling of large-scale data and accelerated training time. The results of hyperparameter tuning demonstrated that increasing the number of epochs led to improved model convergence, higher detection accuracy, and enhanced robust-



ness under diverse environmental conditions. The YOLOv8n model exhibited strong performance in balancing precision and recall, making it well-suited for real-time forest fire detection applications.

## F. Results

1) **Validation Process and Performance Evaluation:** The validation dataset, derived from the Smoke-Fire-Detection-YOLO dataset, consists of unseen images containing fire, smoke, and background scenarios, enabling an unbiased assessment during each training epoch.

At the end of every epoch, the model predictions were evaluated against ground truth annotations to compute key performance metrics such as precision, recall, and mean Average Precision (mAP). Precision reflects the proportion of correctly predicted fire and smoke instances among all detections, while recall measures the ability of the model to identify all actual instances present in the images. The mAP metric, particularly mAP@0.5 and mAP@0.5:0.95, was used to evaluate both classification accuracy and localization performance across multiple IoU thresholds.

The validation results revealed that the model trained for 50 epochs achieved moderate performance, indicating initial convergence and learning of basic feature representations. However, the model trained for 150 epochs demonstrated significantly improved validation metrics, with smoother convergence curves and higher stability. This improvement indicates enhanced feature learning, particularly in detecting small and low-density smoke regions. The extended training also reduced fluctuations in validation loss, suggesting better generalization and reduced overfitting.

2) **Testing Procedure and Final Evaluation:** The testing phase was performed after the completion of training to evaluate the final performance of the model on completely unseen data. Unlike validation, which is used iteratively during training, the testing phase provides a definitive assessment. The test dataset included diverse environmental conditions such as forest landscapes, urban areas, varying lighting conditions, and atmospheric disturbances. The trained YOLOv8n model was applied to this dataset to generate predictions, which were then compared with ground truth annotations to compute performance metrics.

The results showed that the model trained for 150 epochs outperformed the 50-epoch model across all evaluation metrics. It demonstrated higher precision and recall, indicating improved detection accuracy and reduced false positives and false negatives. The model was particularly effective in identifying early-stage smoke, which is critical for forest fire prevention systems. However, some challenges remained in distinguishing smoke from visually similar elements such as fog and clouds, leading to occasional misclassifications. Overall, the testing results confirmed that extended training significantly enhances the model's robustness and generalization capability.

3) **Comparative Analysis of 50 and 150 Epoch Training:** A comparative analysis between the two training configurations highlights the impact of hyperparameter tuning, particu-

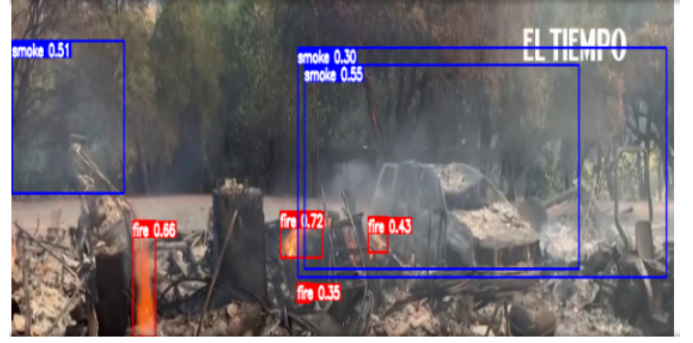


Fig. 3. Bounding Box and the Confidence Score Prediction Of Fire and Smoke

larly the number of epochs, on model performance. The 50-epoch model provided faster training and reasonable baseline results but exhibited limitations in detecting subtle patterns such as thin smoke. In contrast, the 150-epoch model achieved higher accuracy, better localization, and improved consistency across different scenarios.

The extended training allowed the model to learn more complex feature representations, resulting in improved discrimination between fire, smoke, and background classes. This was especially evident in challenging environments where visual ambiguity is high. Additionally, the longer training duration contributed to better convergence, as indicated by reduced validation loss and improved mAP scores.

TABLE I  
TESTING PERFORMANCE COMPARISON OF YOLOV8N FOR 50 AND 150 EPOCHS

| Model Configuration  | Precision | Recall | F1-Score | Accuracy |
|----------------------|-----------|--------|----------|----------|
| YOLOv8n (50 Epochs)  | 0.76      | 0.68   | 0.72     | 0.76     |
| YOLOv8n (150 Epochs) | 0.79      | 0.71   | 0.75     | 0.77     |

These metrics provide a comprehensive evaluation of the model's capability to correctly detect fire and smoke instances while minimizing false detections.

Precision measures the proportion of correctly predicted positive instances among all predicted positives. In the context of fire and smoke detection, it indicates how many detected regions correspond to actual fire or smoke. A high precision value signifies fewer false positives. Mathematically, precision is defined as:

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

where TP represents true positives and FP represents false positives.

Recall evaluates the ability of the model to detect all relevant instances present in the dataset. It measures how effectively the model identifies actual fire and smoke occurrences. High recall indicates fewer missed detections, which is particularly important in safety-critical applications such as forest fire monitoring. Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (10)$$

where FN denotes false negatives.

The F1-score provides a balanced measure by combining precision and recall using their harmonic mean. This metric is particularly useful when there is a need to balance false positives and false negatives. A higher F1-score indicates better overall detection performance. The F1-score is given by:

$$F1 = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \quad (11)$$

Accuracy represents the proportion of correctly predicted instances among all predictions, including both positive and negative classes. It provides a general measure of model correctness but may be less informative in datasets with class imbalance, such as fire detection datasets where background samples dominate. where TN represents true negatives. Accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (12)$$

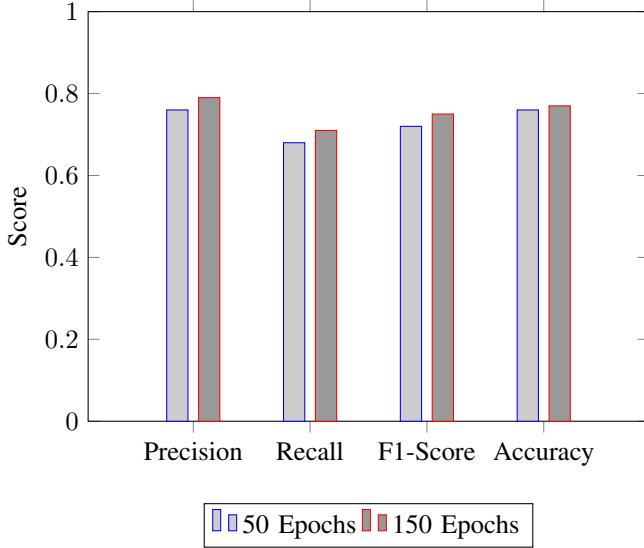


Fig. 4. Performance comparison of YOLOv8n trained for 50 and 150 epochs.

The performance comparison graph illustrates the evaluation of the Ultralytics YOLOv8 nano model (YOLOv8n) trained for 50 and 150 epochs across key metrics, including precision, recall, F1-score, and accuracy, using the Smoke-Fire-Detection-YOLO dataset. The graph clearly shows that the model trained for 150 epochs consistently outperforms the 50-epoch configuration in all evaluation metrics. Specifically, precision improves from 0.76 to 0.79, indicating a reduction in false positive detections, while recall increases from 0.68 to 0.71, demonstrating enhanced capability in identifying actual fire and smoke instances. The F1-score, which balances precision and recall, also shows improvement from 0.72 to

0.75, reflecting a more reliable overall detection performance. Additionally, the slight increase in accuracy highlights better overall prediction correctness. The visual trend confirms that extended training allows the model to learn more robust feature representations, leading to improved generalization and detection performance. This comparison validates the effectiveness of hyperparameter tuning, particularly increasing the number of epochs, in enhancing the reliability of the fire and smoke detection system.

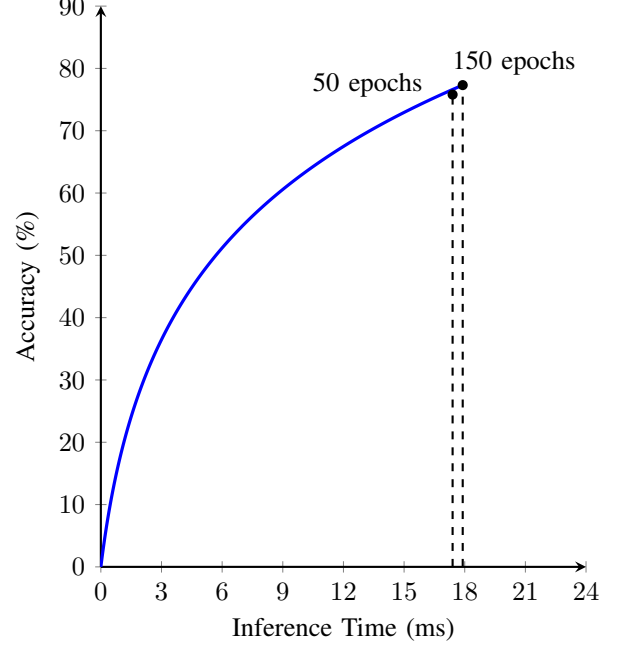


Fig. 5. Inference Time vs Accuracy for Different Training Epochs

This graph illustrates the relationship between inference time and model accuracy for two training configurations: 50 epochs and 150 epochs. The x-axis represents the inference time in milliseconds (ms), while the y-axis represents the model accuracy in percentage.

The logarithmic curve shows that the accuracy increases gradually as the inference time increases. At 17.4 ms, the model trained for 50 epochs achieves an accuracy of 75.80. When the training is extended to 150 epochs, the inference time slightly increases to 17.9 ms, resulting in a higher accuracy of 77.32.

The graph demonstrates a trade-off between inference time and performance: increasing the number of training epochs improves the model accuracy, but also slightly increases the inference time. The smooth logarithmic nature of the curve indicates that the gain in accuracy becomes progressively smaller as inference time increases.

### III. CONCLUSION

This project successfully developed a forest fire and smoke detection system using the YOLOv8n deep learning model for hazard identification. The proposed system utilized image preprocessing, feature extraction, multi-scale feature fusion,

and object detection techniques to accurately identify fire and smoke regions under different environmental conditions. The Smoke-Fire-Detection-YOLO dataset containing over 21,000 annotated instances was used to train and evaluate the model, enabling robust learning of fire and smoke patterns.

Experimental results demonstrated that the YOLOv8n model trained for 50 epochs achieved an accuracy of 0.76 with a training time of 24.135 hours, while the model trained for 150 epochs achieved a higher accuracy of 0.77 with a training time of 74.125 hours. The results indicate that increasing the number of training epochs improves detection performance and enhances the model's capability to identify fire and smoke more effectively. However, the improvement in accuracy is relatively small compared to the substantial increase in computational time, highlighting a trade-off between model performance and training efficiency.

The proposed system demonstrated reliable precision, recall, and overall detection capability, making it suitable for forest monitoring and disaster management applications. Although challenges such as fog, clouds, and varying lighting conditions may still affect detection accuracy, the model showed strong generalization and robustness across different scenarios.

Overall, the study confirms that YOLOv8n provides a scalable, lightweight, and efficient solution for early forest fire and smoke detection. Future enhancements may include integrating thermal imaging, drone-based monitoring, and edge-device optimization to further improve detection accuracy and deployment efficiency in large-scale real-world environments.

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